UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	:	CSC429
EXAMINATION	:	NOVEMBER 2026
TIME	:	

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC429 mock paper (November 2026). Generated answers show working. Real past-paper questions are reproduced with their source reference; no answer is invented where the original scheme is unavailable.

2. Generated marking points state the expected answer and the allocation of marks. Real-bank questions retain their source paper and page reference.

3. Part A = 4 x 5 + 6 x 10; Part B = 1 x 20; total = 100 marks.

4. Generated slots: 2. Real-bank slots: 9. Placeholders: 0. Questions and answers are for local study use only.

5. Structure derived from 20 unique CSC429 past papers; every real question remains traceable to its source page.

FOR MARKERS' USE ONLY

This marking scheme consists of 5 printed pages

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PART A — MARKING GUIDE

QUESTION A1 (5 marks)

Topic: Computer generations and history

Question 1 is based on the following image and statements in Figure 1.
Charles Babbage was an English mathematician, a philosopher and also an inventor. He was born on 1791 in London England. He was known for his contributions to invent machine X to solve four basic arithmetic functions. The machine X has improved its calculations design from Stepped Reckoner machine and early mechanical calculator. The machine X has introduced a number of computing concepts which are still in use today. Although this machine was not actually completed, it is the first machine that deserved to be called as a computer.
Figure 1 : A snippet from biography web page
a) What is the machine X name?
b) In which generation does the machine X introduced?
c) What is the function of machine X?
d) Name the machine that has been invented prior to machine X?
e) What is the title given to Charles Babbage due to his success to invent machine X?

Source: CSC429 (10).pdf, page(s) 2

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A2 (5 marks)

Topic: Instruction formats and addressing modes

Given the following Figure 2.
Instruction
Opcode
Address A
Memory
Operand
Figure 2
Label the addressing technique illustrated in Figure 2.
a)
Explain how the addressing technique in 9(a) is used.
b)
List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A3 (5 marks)

Topic: Parallel and multiprocessor architecture

a)
List THREE (3) characteristics of Symmetric Multiprocessor (SMP).
b)
Describe ONE (1) advantage of Symmetric Multiprocessor (SMP) interms of availability.

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paper and its mark allocation; no answer has been invented.

QUESTION A4 (5 marks)

Topic: Data representation and number systems

Convert 378510 to binary. Show your complete workflow.

Answer: 378510 = 1110110010012

- *1 mark for identifying the target base.*
- *1 mark for a correct repeated-division or place-value method.*
- *1 mark for the correct digit sequence.*
- *1 mark for the correct base subscript.*
- *1 mark for a complete, consistent workflow.*

QUESTION A5 (10 marks)

Topic: Data representation and number systems

Answer both parts for the same expression. Assume an 8-bit byte.
Expression: -60 - (-12)

a) Calculate -60 - (-12) using sign and magnitude.

Answer: left -60 -> 1 0111100; right -12 -> 1 0001100; -60 - (-12) = -48 -> sign-magnitude 1 0110000

- *1 mark for operand representations.*
- *1 mark for sign handling.*
- *1 mark for magnitude arithmetic.*
- *1 mark for the result sign bit.*
- *1 mark for the final byte.*

b) Calculate -60 - (-12) using 1's and 2's complement.

Answer: left -60 -> 1's 11000011, 2's 11000100; right -12 -> 1's 11110011, 2's 11110100; -60 - (-12) = -48 -> 1's 11001111, 2's 11010000

- *1 mark for the positive operand in binary.*
- *1 mark for the 1's complement of the negative operand.*
- *1 mark for adding 1 to form the 2's complement.*
- *1 mark for the correct 8-bit result.*
- *1 mark for converting the result back to decimal.*

QUESTION A6 (10 marks)

Topic: Input/output and interrupts

Question 8 is based on Figure 2.
interrupt
Interrupt
Handler B
H a n d l e r C
Interrupt
Handler A
Interrupt
Interrupt
Handler D Handler E
Figure 2: Multiple interrupts diagram
a) Describe maskable interrupt.
b) What is the sequence of the program execution if the defined priority is implemented in Figure 2?
c) What is the sequence of the program execution if the disable interrupt is implemented in Figure 2?
d) Which of the interruption method between defined priority or disable interrupt can finish Interrupt C earlier? Give your justification.

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

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QUESTION A7 (10 marks)

Topic: Computer architecture and CPU organisation

Describe the specific function of each registers in the CPU.

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A8 (10 marks)

Topic: Computer architecture and CPU organisation

a) Discuss the function of each Central Processing Unit (CPU) component'
b) Elaborate on what will happen in the machine cycle if the bus system is faulty

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A9 (10 marks)

Topic: Input/output and interrupts

a) Differentiate between interrupt-driven 1/0 and programmed 1/0.
b) Discuss ONE (1) benefit of programmed 1/0.
c) Discuss TWO (2) disadvantages of multiple interrupts.

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A10 (10 marks)

Topic: Memory hierarchy and cache

Question I is based on Figure 3.
Computer structure A
Main
Cache L3
cache L2
cache L1
memory
CPU
Computer structure B
Main
cache L3
Cache L4
Cache L2
Cache L1
CPU
memory
Figure 3: Different computer architectures
a) Explain which computer structure can store the data faster compared to the other. Provide relevant justification to support your answer.
b) Identify the fastest cache level and give an appropriate reason.
c) Vvhat is the main benefit for the cache to use SRAM technology?
d) Discuss TWO (2) methods to update cache and main memory contents to ensure data consistency.

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

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PART B — MARKING GUIDE

QUESTION B1 (20 marks)

Topic: Assembly-language programming

Question 1 is based on Figure 2 and Table 1.

y = A -
2C
* D
B + A2

Figure 2

Write an appropriate program using correct assembly language notations to compute the equation shown in Figure 2 using instructions for one-address, two-address, and three-address. The available instructions are provided in Table 1. The register labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register y.

Table 1 : Machine instruction

Three-address	One-address	Two-address
LOADp		
MOV (p,q)		
ADD (p.q.r)		
STORp		
ADD (p,q)		
SUB (p,q,r)		
ADDp		
SUB (p,q)		
MUL (p.q.r)		
SUBp		
MUL (p,q)		
DIV(p,q,r)		
MULp		
DIV(p,q)		
DIVp		

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END OF MARKING SCHEME

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